

OpenSearch

OpenSearch is an **open-source search and analytics suite** derived from Elasticsearch and Kibana, maintained by the OpenSearch community and AWS. It provides a distributed, RESTful search engine and visualization tools for full-text search, log analytics, observability, and security analytics.

Key points

- *Core components:* OpenSearch (search/analytics engine) and OpenSearch Dashboards (visualization UI).
- *Origin and license:* Forked from Elasticsearch/Kibana after licensing changes; fully open-source under the Apache 2.0 license.
- *Architecture:* Distributed, shard-and-replica based, supports horizontal scaling, fault tolerance, and near-real-time indexing.
- *Querying:* RESTful API with JSON DSL for full-text queries, aggregations, filters, and geo/search features; supports SQL querying via a SQL plugin.
- *Indexing and storage:* Inverted-index for full-text, columnar doc-values for aggregations, configurable analyzers, mappings, and replicas.
- **Features:**
 - Search: relevance scoring, multi-field search, fuzzy, phrase, and proximity queries.
 - Analytics: aggregations, rollups, time-series optimizations.
 - Ingest: ingest pipelines, processors, Beats/Logstash and fluentd integrations.
 - Observability: anomaly detection, traces, metrics, logs, and alerting (via plugins).
 - Security: role-based access control, TLS, audit logging (built-in security plugin).
 - Snapshot/restore: backups to S3 and other repositories.
 - Plugins: SQL, alerting, index mgmt., performance analyzer, k-NN for vector search.
- *Vector search:* k-NN plugin for approx. semantic search (nearest neighbor search).
- *Ecosystem & integrations:* Logstash, Fluentd, Prometheus exporters, Grafana, JDBC/ODBC.
- *Deployment:* runs on Linux; deployable on VMs, containers, Kubernetes (operators), and offered as managed services by AWS (OpenSearch Service) and some other providers.
- *Scaling & operations:* supports shard allocation, index lifecycle management (ILM), index templates, and performance tuning for heap, caching, and thread pools.
- *Use cases:* enterprise search, application search, log analytics, security information & event management (SIEM), observability, and semantic/vector search.

When to go for OpenSearch

When an open-source, self-hostable search/analytics platform with strong ecosystem, rich plugins (security, alerting, SQL, k-NN), and control over deployment are needed.

When not to have OpenSearch

When fully managed, minimal-maintenance, cloud-only solutions with extensive auto-scaling and serverless pricing models (some commercial cloud offerings may be a better fit) are needed.

Mark your feedback at the Following Contact Points

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Credits and Gratitude

- Thank all who have mentored, taught and guided. Also, always appreciate who have supported my work.